Dual-Stream Manifold Multiscale Network for Target Recognition in Complex-Valued SAR Image With Electromagnetic Feature Fusion

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***Abstract*—Existing deep-learning-based methods for synthetic aperture radar (SAR) target recognition typically rely solely on the amplitude images without considering the complex characteristic of SAR images, making it difficult to recognize SAR targets with high visual similarity. To solve this issue, a novel dual-stream manifold multiscale network fused with electromagnetic features, i.e., EFMM-Net, is proposed for target recognition in complex-valued SAR images. In EFMM-Net, the attributed scattering center (ASC) model is first used to reconstruct the complex-valued SAR image, thereby highlighting the electromagnetic scattering features of the target. Subsequently, the reconstructed complex-valued SAR image is combined with the original one to construct the dual-stream input. Second, a scattering-guided manifold multiscale (SGMM) backbone is proposed for parallel extraction of data features and electromagnetic scattering features of the target from the dual-stream input. During feature extraction, the SGMM backbone can effectively leverage the phase information of complex-valued SAR images and inject target scattering information into data features through scattering-guided channelwise feature alignment, thus enhancing the awareness of data features to critical scattering characteristics. Finally, to effective fuse the data features and electromagnetic scattering features, a location awareness feature fusion (LAFF) recognition module is proposed. By exploiting coordinate attention, LAFF uses the target location information captured from electromagnetic scattering features to direct the feature fusion process, thereby increasing the focus of fusion features on the target region. The extensive recognition results of three-class and six-class ship targets in the OpenSARShip 2.0 dataset demonstrate the effectiveness and superiority of the proposed method.**

***Index Terms*—Attributed scattering center (ASC) model, feature fusion, manifold network, ship target recognition, synthetic aperture radar (SAR).**

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I. Introduction

As an active earth observation system, synthetic aperture radar (SAR) has the advantages of being able to acquire high-resolution images of the observed scenes under all-day and all-weather observation conditions [1], [2]. In recent years, SAR has been widely used in remote sensing image target recognition [3], [4], target detection [5], [6], semantic segmentation [7], [8], and other fields. Among them, SAR target recognition has become one of the research hotspots in the field of remote sensing. However, SAR imaging results can be easily affected by wavelength, incidence angle, polarization type, and other factors. Besides, SAR image characteristics, such as scattering center anisotropy and contour blurring induced by special imaging mechanism, differentiate SAR images significantly from optical images, making SAR target recognition more difficult.

In early years, SAR target recognition algorithms usually need to extract target features manually. Commonly used features include image domain features, transform domain features, and electromagnetic scattering features. Typical image domain features contain target shapes and contours [9], histogram of orientation gradient (HOG) [10], etc. For transform domain features, independent component analysis (ICA) [11], Fisher linear discriminant analysis (LDA) [12], and their variants usually transform SAR image features into feature vectors, and then recognize SAR targets based on signal analysis methods. The electromagnetic scattering features can effectively express the strong scattering centers of SAR targets, thus reflecting the target geometry, structure and location information [13]. Although these artificially designed features are more consistent with SAR image characteristics, they need to rely on the expert experience and prior knowledge, and the extracted features have limited representation ability and poor generalization performance.

Deep learning models can automatically learn deep embedding features of images end-to-end, so they have gained wide applications in the field of SAR target recognition in recent

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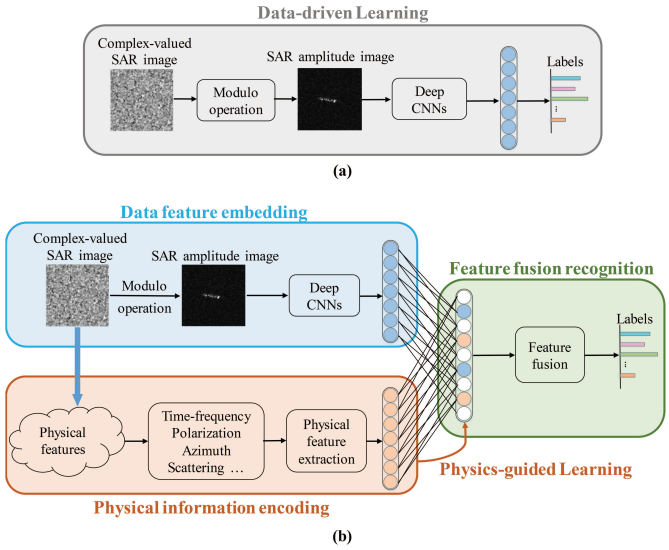


Fig. 1. Two mainstream deep-learning-based SAR target recognition paradigms. (a) Data-driven learning. (b) Physics-guided learning.

years. At present, deep-learning-based SAR target recognition methods mainly fall into two categories: data-driven learning and physics-guided learning.

A shown in Fig. 1(a), the data-driven learning methods mainly focus on SAR amplitude images. These approaches generally first expand the training data through data augmentation techniques, and then use convolutional neural networks (CNNs) to extract shape, texture, and other visual features of SAR targets [14]. Considering the complicated structure of deep CNN models in the field of optical image processing, Chen et al. [15] proposed a lightweight all-convolutional network (A-ConvNets), which discards the fully connected (FC) layer with a large number of parameters, and thus is more suitable for SAR target recognition with a small sample size. Due to the manual operations of SAR image annotation, training sets often consist of only a limited number of labeled samples. To address this, Pei et al. [16] introduced contrastive learning to SAR target recognition and improved the target recognition performance even under the condition of fewer labeled samples. For ship targets with large intraclass variability and interclass similarity, He et al. [17] proposed a multitask learning framework to constrain the optimization process of triplet loss and softmax loss using Fisher discrimination regularized metric learning. Moreover, recent SAR target recognition studies also incorporate modules that have good performance in optical images. For example, the long short-term memory (LSTM) network is used for learning sequential SAR image features [18], the vision transformer (ViT) is introduced for extracting global features [19], and the attention mechanism is adopted for capturing more discriminative target features [20]. Although the data-driven methods can learn visual features and data distribution properties of SAR amplitude images, the special electromagnetic scattering imaging mechanism of SAR makes it difficult to visually distinguish targets with complex scattering. Therefore, relying solely on SAR amplitude image information cannot provide sufficient discriminable features for SAR target recognition.

Recently, scholars have proposed a physics-guided SAR target recognition framework shown as Fig. 1(b), which incorporates the physical features of complex-valued SAR images, such as time–frequency, polarization, azimuth, scattering, and other features into deep learning models, prompting the models to learn the unique physical features of SAR targets. Huang et al. [21] introduced the hierarchical frequency features into SAR target recognition and proposed a spatial–frequency-domain feature fusion method. Xie et al. [22] proposed a double-branch ship target classification network to extract polarization and texture features of dual-polarized ship targets. Oh et al. [23] regarded the pose angle of SAR target estimated by the pose angle probability estimation network as the prior information and incorporated it into SAR target recognition. Wang et al. [24] used the graph attention network to estimate the azimuth angles of SAR images, thus assisting the model to aggregate the spatial features of sequential SAR images. Taking advantages of electromagnetic scattering features, Zhang et al. [25] groundbreakingly used bag of visual words (BOVW) to transform electromagnetic features into feature vectors. The vectors are then fused with data features extracted by real-valued network to achieve SAR target recognition. Li et al. [26] decomposed the target into different components according to the geometric scattering types of attributed scattering centers (ASCs), and then combined these component features with global features to obtain more robust target feature representations. Zhang et al. [27] proposed a multiscale global scattering feature association network, which fully uses the scattering features and high-dimensional deep features of SAR ship targets. Liu et al. [28] and Feng et al. [29], [30] also conducted some related works by incorporating electromagnetic scattering features and achieved notable results in SAR target recognition. Liao et al. [31] extracted the main scattering centers of the target using the deep unfolding network, thus avoiding the mismatch between scattering center features and classifier. It is evident that unlike data-driven learning approaches, physics-guided SAR target recognition models are able to intelligently use the physical characteristics of the target to guide deep networks for autonomous learning. As a result, the prediction outcomes are more consistent with the physical cognition of SAR images.

Despite the success of the aforementioned methods in using electromagnetic features for SAR target recognition, most of them still suffer from the following two issues.

1) *The Backbone Network for Feature Extraction Does Not Consider the Phase Information:* SAR images are usually in complex-valued form and contain both the amplitude and phase. The amplitude represents the backscattering intensity of the target, and the phase reflects the contour and reflectivity of the target. However, the backbone networks of the existing SAR target recognition methods are typically real-valued CNNs. These networks only use SAR amplitude images while ignoring phase information, resulting in limited recognition performance.
2) *Specific Electromagnetic Feature Extractors Need To Be Carefully Designed:* Due to the discrepancy between electromagnetic scattering features and data features of the target, the existing approaches typically require

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careful design of specific electromagnetic feature extractors, as well as the conversion of electromagnetic features into a form that can be easily fused with the data features extracted by deep networks. Although these methods perform well, they limit the network from further extracting deep electromagnetic features.

To mine the phase information, Zeng et al. [32] and Zhu et al. [33] proposed different complex-valued network architectures for SAR target recognition, respectively. These networks treat complex data as two separate channels consisting of real and imaginary parts, destroying the covariant relationship between them. Distinct from the complex-valued networks above, Chakraborty et al. [34], [35] extended complex-valued deep learning to Riemannian manifolds and constructed a series of manifold network models. The proposed SurReal [34] model emphasizes the preservation of intrinsic geometry of complex data. Liu and Lang [36] designed a multimanifold feature fusion (MMFF) method, which considers both the underlying geometry and the multimodal information of complex-valued SAR images. Compared with the traditional SAR target recognition methods based on complex-valued networks, the manifold network can effectively use the geometric property of complex data.

Combining the advantages of manifold networks in processing complex data and the electromagnetic scattering characteristics of SAR targets, a novel dual-stream manifold multiscale SAR target recognition network fused with electromagnetic features, i.e., EFMM-Net, is proposed in this article. First, we restore the echo data of target according to the ASC model and reconstruct the complex-valued SAR image characterizing the electromagnetic scattering characteristics of the target. Then, the original complex-valued SAR image is combined with the reconstructed one to construct the dual-stream input. Second, to effectively use the phase information and scattering characteristics of the dual-stream complex-valued SAR images, we propose a scattering-guided manifold multiscale (SGMM) backbone. During the feature extraction process, the SGMM backbone injects scattering information into the data features through channelwise feature alignment, thereby enhancing the awareness of data features to target scattering characteristics. Finally, for effective fusion of the two types of features, we design a location awareness feature fusion (LAFF) recognition module. The LAFF adopts coordinate attention to direct the fusion features to focus on the target region, thereby further improving SAR target recognition accuracy. The main contributions of this work are summarized as follows.

1) A novel dual-stream network fused with electromagnetic features, termed EFMM-Net, is proposed for target recognition in complex-valued SAR images. This method uses two identical manifold multiscale networks to simultaneously extract data features and electromagnetic scattering features of the target from dual-stream complex-valued SAR images, thus avoiding the additional design of specific electromagnetic feature extractors. Significantly, the complex-valued SAR images reconstructed using the ASC model effectively enhance the target structural feature representations.
2) To enhance the awareness of data features to target scattering characteristics, a SGMM backbone is designed to inject transferable scattering information into data features through scattering-guided channelwise feature alignment. Furthermore, the manifold multiscale network can effectively leverage the phase information of complex-valued SAR images while preserving the geometric properties of complex data.
3) An LAFF recognition module is introduced to achieve effective fusion of the two types of features and SAR target recognition. By exploiting coordinate attention, LAFF can capture the target location information from electromagnetic scattering features and drive the fusion features to focus more on the target region, thereby improving the SAR target recognition accuracy.

The rest of this article is organized as follows. Section II introduces the preliminary knowledge of manifold network. Section III describes the specific details of the proposed EFMM-Net method, including the dual-stream complex-valued SAR image input module, the SGMM backbone, and the LAFF recognition module. In Section IV, extensive experiments are conducted on the OpenSARShip 2.0 dataset to demonstrate the effectiveness and superiority of the proposed method. Finally, Section V concludes the whole article and discusses the future work.

II. Preliminary Knowledge of Manifold Network

Generally, a nonzero complex number can be expressed as $z = x + j \cdot y = |z|e^{j\cdot\theta} \in \widetilde{\mathbb{C}}$, where $j = (-1)^{1/2}$ is the imaginary unit, and $\widetilde{\mathbb{C}}$ represents the nonzero complex plane. In the Cartesian coordinates, x and y correspond to the real and imaginary parts of the complex number, respectively. In the polar coordinates, $|z|$ and θ indicate its amplitude and phase, respectively. According to the polar coordinate representation, we can regard points on the nonzero complex plane as points on the scaling–rotation product manifold, thereby mapping complex numbers from the original nonzero complex plane to the product manifold.

The weighted average of M points $\{z_i = |z_i|e^{j\cdot\theta_i}\}_{i=1}^{M}$ on a Riemannian manifold is called the weighted Fréchet mean (wFM) [34]. We define the complex convolution as wFM on the scaling–rotation product manifold for complex numbers, which is calculated as

$$\mathrm{wFM}\left(\{z_i\},\{w_i\}\right) := \arg\min_{z'\in\widetilde{\mathbb{C}}} \sum_{i=1}^{M} w_i d^2\left(z_i, z'\right) \tag{1}$$

$$d\left(z_a, z_b\right) = \sqrt{\log^2\frac{|z_b|}{|z_a|} + \left\|\mathrm{logm}\left(R\left(\theta_b\right)R\left(\theta_a\right)^{-1}\right)\right\|^2} \tag{2}$$

$$R\left(\theta\right) = \begin{bmatrix} \cos\left(\theta\right) & -\sin\left(\theta\right) \\ \sin\left(\theta\right) & \cos\left(\theta\right) \end{bmatrix} \tag{3}$$

where $\{w_i\}_{i=1}^{M}$ is the weight that satisfies $\sum_{i=1}^{M} w_i = 1$, d denotes the manifold distance, and logm represents the matrix logarithm. The wFM convolution is equivariant to complex-valued scaling by preserving the geometric property of complex data, so the amplitude and phase of complex data can be effectively used.

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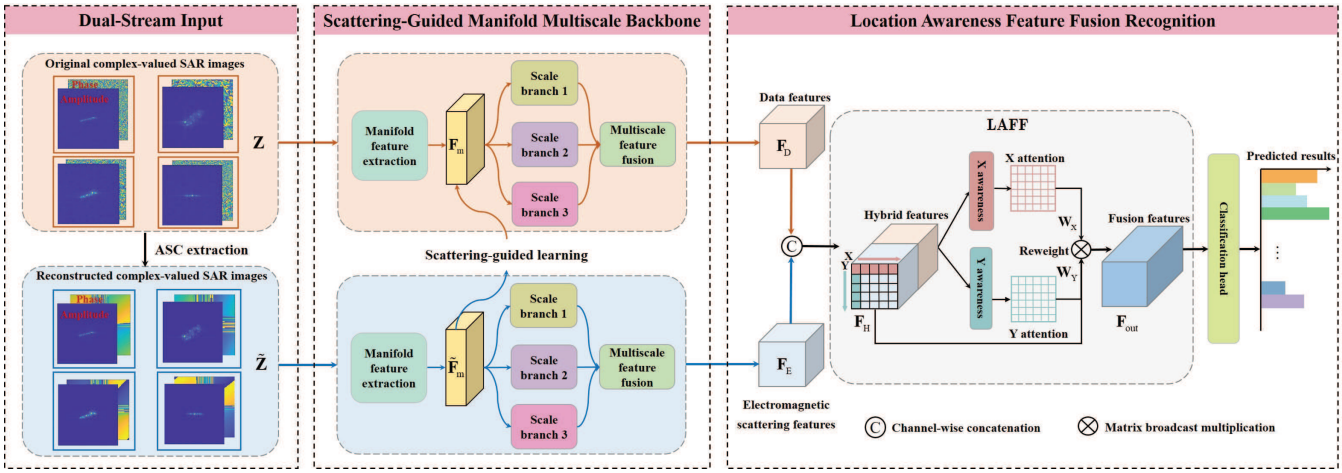


Fig. 2. Overall structure of the proposed EFMM-Net.

Nonlinear activation in the manifold space can be realized by applying ReLU activation function in its tangent space, called tReLU. The tReLU activation consists of three steps: 1) apply logarithm operations to map points from the product manifold space to its tangent space; 2) perform ReLU activation in the tangent space; and 3) apply exponential operations to return from the tangent space to the product manifold space. For a complex number z_i, the three-step tReLU activation operation can be summarized as

$$|z_i| \overset{\text{tReLU}}{\rightarrow} \exp\left(\text{ReLU}\left(\log\left(|z_i|\right)\right)\right) = \max\left(|z_i|, 1\right) \quad (4)$$

$$R\left(\theta_i\right) \overset{\text{tReLU}}{\rightarrow} \text{expm}\left(\text{ReLU}\left(\text{logm}\left(R\left(\theta_i\right)\right)\right)\right) = R\left(\max\left(\theta_i, 0\right)\right) \quad (5)$$

where expm represents the matrix exponent.

For the manifold network, the FC layer transforms complex features to the real domain through distance transformation, thereby enabling further feature extraction and optimization. Suppose that the input feature maps of the FC layer consist of C channels with K pixels per channel, i.e., there are a total of $C \times K = N$ complex numbers, denoted as $\{z_i\}_{i=1}^{N}$. Given N weights $\{w_i\}_{i=1}^{N}$ for these individual complex numbers, the distance transformation first calculates their mean $\text{wFM}(\{z_i\}, \{w_i\})$ and then computes the manifold distance l_i from z_i to the mean $\text{wFM}(\{z_i\}, \{w_i\})$

$$l_i = d\left(z_i, \text{wFM}\left(\{z_i\}, \{w_i\}\right)\right). \quad (6)$$

The output features of FC layer are real-valued and of the same size as the input features.

III. EFMM-Net

In this section, the overall structure of EFMM-Net is first presented. Then, the three main modules and loss function are elaborated, respectively.

A. Overall Structure

The overall structure of EFMM-Net is illustrated in Fig. 2, which consists of three main modules: the dual-stream complex-valued SAR image input module, the SGMM backbone module, and the LAFF recognition module. We first use the ASC model to extract the scattering centers of the target to obtain the reconstructed complex-valued SAR image, and then it is combined with the original one to form the dual-stream input. In the SGMM backbone, two identical manifold multiscale networks are adopted to process the dual-stream input to extract target data features and electromagnetic scattering features. Furthermore, a scattering-guided learning method is introduced, which aims to inject target scattering information into the data features, thereby enhancing the awareness of data features to target scattering characteristics. Within the LAFF recognition module, the coordinate attention is used to enhance the feature fusion process, promoting the fusion features to focus more on the target region. The final SAR target recognition results can be obtained by classifying the fusion features with the classification head.

B. Dual-Stream Complex-Valued SAR Image Input

In SAR imaging researches, the ASC model can characterize the response of each scattering center of the target as a function of range frequency and azimuth angle. The inherent geometrical and physical attributes contained in the model parameters can accurately describe the electromagnetic scattering characteristics of the target. In view of this, we apply a simplified ASC model to extract target scattering centers, which are then used to reconstruct the complex-valued SAR image.

In the high-frequency region, the echo data of the original complex-valued SAR image $\mathbf{Z}$ can be modeled as the sum of the responses from a series scattering centers of the target, which can be expressed as

$$E\left(f, \varphi; \Theta\right) = \sum_{q=1}^{Q} E_q\left(f, \varphi; \Theta_q\right) + n\left(f, \varphi\right) \quad (7)$$

where $E(f, \varphi; \Theta)$ represents the total echo data, $E_q(f, \varphi; \Theta_q)$ denotes the echo data of the scattering center q, and $n(f, \varphi)$ represents the additive noise and clutter. f and φ are the range frequency and azimuth angle, respectively, and $\Theta = \{\Theta_q\}(q = 1, 2, \ldots, Q)$ are the parameter sets of the

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scattering centers. For a single scattering center, the ASC model can be described as

$$E_q(f,\varphi;\Theta_q) = A_q \cdot \left(j\frac{f}{f_c}\right)^{\alpha_q} \cdot \operatorname{sinc}\left(2\pi\frac{f}{c}L_q \sin(\varphi-\bar{\varphi}_{0q})\right) \cdot \exp\left[-j4\pi\frac{f}{c}\left(x_q\cos\varphi + y_q\sin\varphi\right)\right] \cdot \exp\left(-j2\pi f\gamma_q\sin\varphi\right) \quad (8)$$

where $\Theta_q = \{A_q, \alpha_q, x_q, y_q, L_q, \bar{\varphi}_{0q}, \gamma_q\}$ is the parameter set of the scattering center q. f_c is the radar center frequency, and c is the propagation velocity of the electromagnetic wave. In the parameter set Θ_q, A_q is the complex amplitude, α_q is the frequency dependence factor, and x_q and y_q are the position coordinates in the range and azimuth dimensions, respectively. In addition, $\{L_q, \bar{\varphi}_{0q}, \gamma_q\}$ are the azimuth dependence factors. For the distributed ASC that satisfies $\gamma_q = 0$, the parameters L_q and $\bar{\varphi}_{0q}$ denote its length and initial orientation angle, respectively. For the localized ASC, the parameters satisfy $L_q = \bar{\varphi}_{0q} = 0$. When the ratio of the radar operating bandwidth to the center frequency is small, the parameter α_q makes little effect. Meanwhile, in SAR imaging systems, γ_q is usually a very small value, so terms related to α_q and γ_q can be neglected. In this case, the ASC model can be simplified as [25]

$$E_{sq}(f,\varphi;\Theta_q) = A_q \cdot \operatorname{sinc}\left(2\pi\frac{f}{c}L_q\sin(\varphi-\bar{\varphi}_{0q})\right) \cdot \exp\left[-j4\pi\frac{f}{c}\left(x_q\cos\varphi + y_q\sin\varphi\right)\right]. \quad (9)$$

In the simplified ASC model, the extraction of the scattering centers becomes the estimation of parameters $\Theta_q = \{A_q, x_q, y_q, L_q, \bar{\varphi}_{0q}\}$. We use the modified RELAX method to estimate the parameter set $\hat{\Theta}_q$, and the detailed optimization procedure can be found in [37]. By substituting the estimated parameter set $\hat{\Theta}_q = \{\hat{A}_q, \hat{x}_q, \hat{y}_q, \hat{L}_q, \hat{\bar{\varphi}}_{0q}\}$ into (9), the reconstructed echo data of the scattering center q can be obtained

$$\hat{E}_{sq}(f,\varphi;\hat{\Theta}_q) = \hat{A}_q \cdot \operatorname{sinc}\left(2\pi\frac{f}{c}\hat{L}_q\sin(\varphi-\hat{\bar{\varphi}}_{0q})\right) \cdot \exp\left[-j4\pi\frac{f}{c}\left(\hat{x}_q\cos\varphi + \hat{y}_q\sin\varphi\right)\right]. \quad (10)$$

It is assumed that there are totally Q scattering centers, the reconstructed echo data can be expressed as

$$E_R(f,\varphi;\hat{\Theta}) = \sum_{q=1}^{Q}\hat{E}_{sq}(f,\varphi;\hat{\Theta}_q). \quad (11)$$

Then the reconstructed complex-valued SAR image $\tilde{\mathbf{Z}}$ can be obtained after applying SAR processing to the reconstructed echo data E_R.

Fig. 3 shows the original and the reconstructed complex-valued SAR images of the measured Container Ship target in the OpenSARShip 2.0 dataset. As shown in Fig. 3(a), the original SAR amplitude image is interfered by background noise and sea clutter. While in the reconstructed SAR amplitude image shown as Fig. 3(c), the noise and sea clutter are suppressed, making the location and structure of ship target more visible. Moreover, we calculate the target-to-background ratio (TBR) [38] of the original and reconstructed

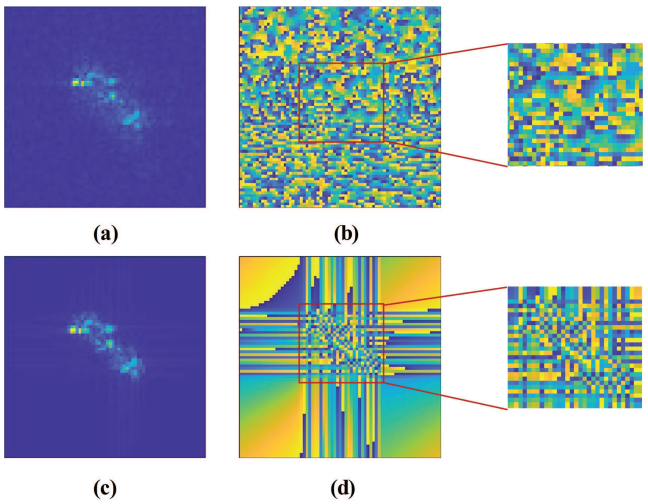


Fig. 3. Original and reconstructed complex-valued SAR images of Container Ship target in the OpenSARShip 2.0 dataset. (a) and (b) Amplitude and phase of the original SAR image, respectively. (c) and (d) Corresponding reconstruction results using the simplified ASC model. The number of extracted scattering centers is set to 40.

SAR amplitude images, and their TBR values are 15.29 and 28.18 dB, respectively. It can be seen that the TBR value of the reconstructed SAR amplitude image increases by about 13 dB, indicating that the application of simplified ASC model can effectively suppress background noise and sea clutter. As seen in Fig. 3(b), it can be found that the original phase image approximately obeys the uniform distribution. However, in the reconstructed phase image shown in Fig. 3(d), due to the sinc response characteristics of SAR imaging, the energy is mainly concentrated in the target region, i.e., the red box area, and the target contour can be observed. Overall, the original complex-valued SAR image contains the data features of the target, and the reconstructed complex-valued SAR image reflects the structural electromagnetic scattering features, such as target location, structure, and contour. Therefore, we use the reconstructed complex-valued SAR image together with the original one as the dual-stream input of EFMM-Net.

C. SGMM Backbone

Previous target recognition models predominantly use real-valued CNNs as backbone networks to extract visual features from SAR amplitude images, while neglecting phase information. Traditional complex-valued deep networks destroy the underlying geometry of complex-valued SAR images, resulting in extracted features cannot reflect target geometric characteristics. To address these limitations, we construct an SGMM backbone, which uses two identical manifold multiscale networks to extract data features and electromagnetic scattering features of the target in parallel. Due to the complementarity and transferability of these two types of features, we further introduce scattering-guided learning in the SGMM backbone. This method injects the transferable scattering information into the data feature representations through channelwise feature alignment, thereby enhancing the awareness of data features to target scattering characteristics and reducing the interference of background noise and clutter.

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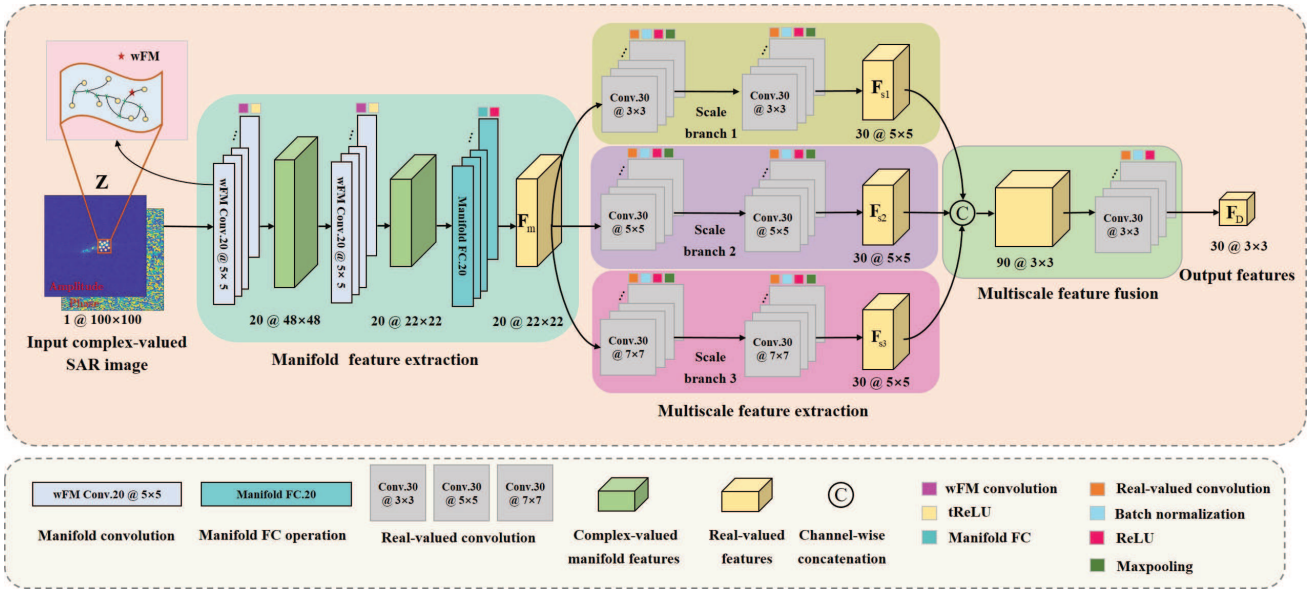


Fig. 4. Architecture of the proposed manifold multiscale network, where "C@H × W" indicates that the complex-valued manifold features or the real-valued features have C channels and the feature map size is H × W.

In the following, we will introduce the composition of manifold multiscale network and the principle of scattering-guided learning method.

1) Manifold Multiscale Network: As seen in Fig. 4, the manifold multiscale network consists of a shallow manifold feature extraction part and a deep multiscale feature extraction and fusion part. The former makes full use of phase information while preserving the geometric properties of complex-valued SAR data. The latter uses three different sizes of convolution kernels to further extract multiscale features, thereby enhancing the feature representation capability of the target. Specifically, the shallow extractor contains two wFM convolution layers and a manifold FC layer. The notation "wFM Conv.20@5 × 5" represents a wFM convolution layer containing 20 5 × 5 kernels with tReLU activation, while "Manifold FC.20" denotes a manifold FC layer with 20 output channels. Given the original complex-valued SAR image $\mathbf{Z}$ and its reconstructed counterpart $\tilde{\mathbf{Z}}$, after processing by their respective shallow extractors, we can obtain the corresponding intermediate manifold features $\mathbf{F}_m$ and $\tilde{\mathbf{F}}_m$. These features are transformed into real-valued representations, thus contributing to the subsequent realization of scattering-guided channelwise feature alignment and deep multiscale feature extraction.

Generally, larger convolutional kernels can capture broader contextual information by extending the receptive fields, whereas smaller kernels excel at preserving local structural details. To leverage the complementary benefits of varying convolutional kernels, three scale branches with different kernel sizes are adopted in the deep multiscale feature extraction and fusion part to further extract multiscale features of the target. As shown in Fig. 4, each branch comprises two convolution layers with the same kernel size, and each convolution layer is successively composed of convolution, batch normalization, ReLU, and maxpooling. The notation "Conv.30@K×K" specifies a convolution layer with 30 output channels and the kernel size is K × K. Let $\mathbf{F}_{\mathrm{s}i}, i = 1, 2, 3$ denote the output features of the three branches. To comprehensively use multiscale features, we first concatenate them along the channel dimension, and then use a 3 × 3 convolution layer to achieve multiscale feature interaction and channel compression, thereby obtaining the multiscale feature fusion output.

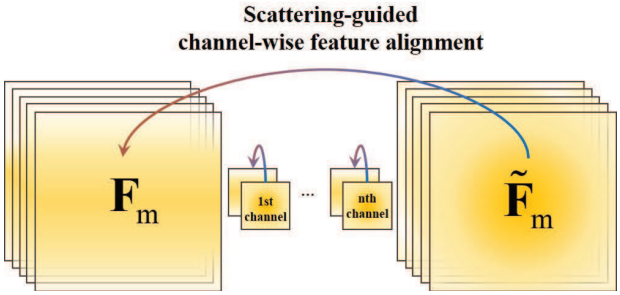


Fig. 5. Principle of the scattering-guided learning method.

2) Scattering-Guided Learning: The data features of original complex-valued SAR images provide the global contextual information including both target and background, whereas the reconstructed complex-valued SAR images reveal the electromagnetic scattering characteristics of the target. To leverage this complementary relationship, we further propose a scattering-guided learning method to enhance the awareness of the data features to target scattering characteristics. As shown in Fig. 5, inspired by feature-based knowledge distillation [39], [40], we inject the target electromagnetic scattering information contained in manifold features $\tilde{\mathbf{F}}_m$ into $\mathbf{F}_m$ through channelwise feature alignment. During the feature alignment process, we adopt a probability map-based feature alignment approach instead of enforcing pixel-by-pixel feature matching, thereby avoiding the alignment difficulty caused by the distribution differences of some pixels in $\tilde{\mathbf{F}}_m$ and $\mathbf{F}_m$. The two features $\mathbf{F}_m$ and $\tilde{\mathbf{F}}_m$ are first normalized by a channelwise softmax function to generate corresponding

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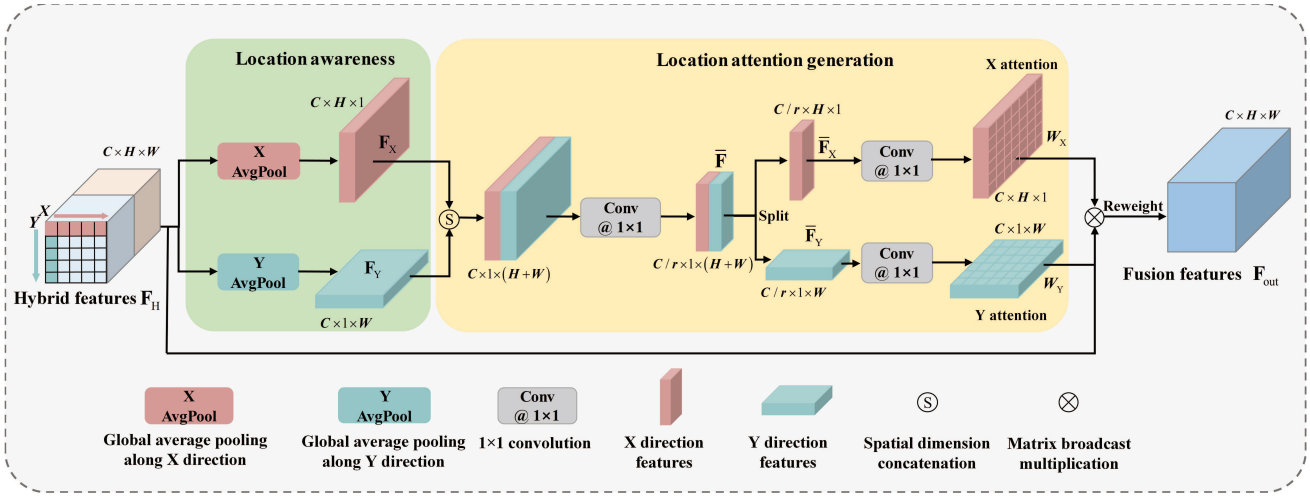


Fig. 6. Network architecture of the LAFF.

cross-channel probability maps. In particular, the calculation of channelwise normalization is formulated as

$$\mathcal{N}\left(\mathbf{F}_m(c,i)\right) = \frac{\exp\left(\mathbf{F}_m(c,i)\right)}{\sum_{i=1}^{H\times W}\exp\left(\mathbf{F}_m(c,i)\right)} \tag{12}$$

$$\mathcal{N}\left(\tilde{\mathbf{F}}_m(c,i)\right) = \frac{\exp\left(\tilde{\mathbf{F}}_m(c,i)\right)}{\sum_{i=1}^{H\times W}\exp\left(\tilde{\mathbf{F}}_m(c,i)\right)} \tag{13}$$

where $\mathbf{F}_m(c,i)$ and $\tilde{\mathbf{F}}_m(c,i)$ denote the pixel values at the ith spatial position of the cth channel in $\mathbf{F}_m$ and $\tilde{\mathbf{F}}_m$, respectively, and $\mathcal{N}(\mathbf{F}_m)$ and $\mathcal{N}(\tilde{\mathbf{F}}_m)$ represent their channelwise normalized probability maps. Next, the feature distribution discrepancy between normalized probability maps is quantified via Kullback–Leibler (KL) divergence, and this metric is leveraged as the channelwise feature alignment loss for subsequent optimization of EFMM-Net, which is given by

$$\begin{aligned} L_{\mathrm{CN}} &= KL\left(\mathbf{F}_m \,\|\, \tilde{\mathbf{F}}_m\right) \\ &= \frac{1}{C}\sum_{c=1}^{C}\sum_{i=1}^{H\times W}\mathcal{N}\left(\mathbf{F}_m(c,i)\right)\cdot\log\left[\frac{\mathcal{N}\left(\mathbf{F}_m(c,i)\right)}{\mathcal{N}\left(\tilde{\mathbf{F}}_m(c,i)\right)}\right]. \end{aligned} \tag{14}$$

Since the probability map-based feature alignment approach mainly focuses on aligning significant target scattering features between $\tilde{\mathbf{F}}_m$ and $\mathbf{F}_m$, it ensures stable network training even if some of the pixels in $\tilde{\mathbf{F}}_m$ and $\mathbf{F}_m$ differ significantly. In addition, although this process may attenuate the background information irrelevant to recognition, the injection of electromagnetic scattering information from $\tilde{\mathbf{F}}_m$ into $\mathbf{F}_m$ enhances beneficial scattering features critical for recognition within $\mathbf{F}_m$, thereby contributing to improve the final recognition accuracy.

D. LAFF Recognition

Following feature extraction through the SGMM backbone, both the original and reconstructed complex-valued SAR images are transformed into corresponding data features $\mathbf{F}_D$ and electromagnetic scattering features $\mathbf{F}_E$. Since the original complex-valued SAR image is affected by the background noise and clutter, the extracted data features contain both the target features and other interference features. In contrast, the extracted electromagnetic scattering features are more focused on the target region, weakening the influence of interference features. To leverage these advantages, we develop an LAFF recognition module that uses scattering features to guide the feature fusion process, thereby directing the fusion features to focus on the target region.

The network architecture of LAFF is shown as Fig. 6, where the hybrid features $\mathbf{F}_H \in \mathbb{R}^{C\times H\times W}$ are obtained by concatenating the data features $\mathbf{F}_D$ and the electromagnetic scattering features $\mathbf{F}_E$ along the channel dimension. In the feature fusion process, LAFF first perceives the target location from the hybrid features using two global average pooling (AvgPool) kernels along the horizontal and vertical directions, respectively, and generates two 1-D pooled features $\mathbf{F}_X \in \mathbb{R}^{C\times H\times 1}$ and $\mathbf{F}_Y \in \mathbb{R}^{C\times 1\times W}$. Since the two AvgPool kernels operate independently, LAFF can aggregate target location information along one direction while preserving location information along the other direction. Such operations enable the model to accurately perceive the spatial location of the target in the hybrid features.

To generate the target location attention weights, the two 1-D pooled features are first concatenated along the spatial dimension and then passed through a 1×1 convolution layer for channel compression. Similar to the squeeze-and-excitation network (SENet) [41], the adopted coordinate attention uses a channel compression ratio r to reduce the number of channels, thereby decreasing the computational cost [42]. In our experiments, we observe that LAFF achieves optimal performance when $r = 6$. Afterward, we split the compressed features $\bar{\mathbf{F}}$ along the spatial dimension to obtain two separated features $\bar{\mathbf{F}}_X \in \mathbb{R}^{C/r\times H\times 1}$ and $\bar{\mathbf{F}}_Y \in \mathbb{R}^{C/r\times 1\times W}$. Finally, by means of two additional 1×1 convolutions, $\bar{\mathbf{F}}_X$ and $\bar{\mathbf{F}}_Y$ are transformed into new features with the same number of channels as the hybrid feature $\mathbf{F}_H$, namely, the target location attention weights $\mathbf{W}_X \in \mathbb{R}^{C\times H\times 1}$ and $\mathbf{W}_Y \in \mathbb{R}^{C\times 1\times W}$ along the horizontal and vertical directions.

By reweighting hybrid features $\mathbf{F}_H$ using the two location attention weights $\mathbf{W}_X$ and $\mathbf{W}_Y$, the feature fusion output can be obtained. Compared with $\mathbf{F}_H$, the fusion features $\mathbf{F}_{\mathrm{out}}$ can effectively focus on the target region while reducing the impact of background noise and clutter.

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We use the classification head to perform SAR target recognition on the fusion features. The classification head contains of $T3 \times 3$ convolutional kernels, where T represents the number of target categories. Then the BN and softmax functions are applied to generate the final recognition results.

E. Loss Function

The EFMM-Net is optimized using two kinds of loss functions, i.e., the standard cross-entropy loss L_{CE} and the channelwise feature alignment loss L_{CN}. First, we use L_{CE} to supervise the entire optimization process, which is formulated as

$$L_{\mathrm{CE}} = -\frac{1}{K}\sum_{k=1}^{K}\sum_{t=1}^{T} y_{kt}\log\left(\hat{y}_{kt}\right) \tag{15}$$

where K represents the number of samples, y_{kt} denotes the true label of the kth group of original and reconstructed complex-valued SAR image samples belonging to the tth category, and $\hat{y}_{kt}$ indicates the prediction result.

Besides the cross-entropy loss, we also propose a channelwise feature alignment loss L_{CN} in Section III-C. As an auxiliary loss term, L_{CN} aims to motivate the SGMM backbone to extract discriminative features reflecting target scattering characteristics during the network optimization process. In summary, the joint loss L of the proposed EFMM-Net is given below

$$L = L_{\mathrm{CE}} + \lambda \cdot L_{\mathrm{CN}} \tag{16}$$

where λ is the weight coefficient of L_{CN}, which is set to 0.5 in this article. Assigning a relatively small weight coefficient to L_{CN} helps smooth the alignment loss gradients while preventing the feature alignment process from interfering with the main recognition task, thereby enhancing the training stability. Furthermore, the weighting operation makes L_{CN} only serve as an auxiliary loss term, and the main network optimization is still dominated by cross-entropy loss corresponding to the category labels, thereby ensuring that the aligned features retain the key information useful for recognition.

IV. Experimental Results and Analysis

In this section, experiments are conducted to verify the performance of the proposed EFMM-Net for SAR target recognition. Section IV-A introduces the OpenSARShip 2.0 dataset used for SAR target recognition. Section IV-B describes the experimental settings and evaluation metrics. Section IV-C provides the recognition results and compares them with other SAR target recognition methods to verify the superiority of the proposed method. Section IV-D evaluates the SAR target recognition performance under different polarizations. Section IV-E assesses the effectiveness of main modules of EFMM-Net through ablation studies. In addition, Section IV-F introduces the SAR target recognition performance of EFMM-Net under complex conditions, including limited training sample and noise interference. Finally, Section IV-G discusses the effectiveness of scattering-guided learning method through visualization analysis.

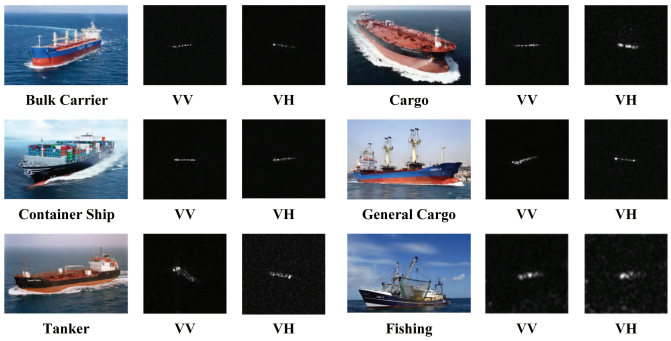


Fig. 7. Optical images and SAR images of ship targets in the OpenSARShip 2.0 dataset. VV and VH represent SAR images of ship targets corresponding to different polarizations in the SLC data.

TABLE I
Number of Samples for Each Category of Ship Target

Class	Bulk Carrier	Container Ship	Tanker
Number	569	435	476
Class	Cargo	General Cargo	Fishing
Number	419	257	268

A. Dataset Description

The OpenSARShip 2.0 is a public SAR ship target dataset [43], [44], which is actually collected from Sentinel-1 images under different environment conditions. In this article, single-look complex (SLC) images under VV and VH polarizations in the OpenSARShip 2.0 dataset are used to evaluate the SAR target recognition performance of the proposed EFMM-Net. Fig. 7 shows the optical images and SAR images under different polarizations of ship targets. Following the ship target classification dataset setup in [22], we select 1480 ship chips from Bulk Carrier, Container Ship, and Tanker targets to construct the three-class ship target dataset, and 2424 ship chips from Bulk Carrier, Container Ship, Tanker, Cargo, General Cargo, and Fishing targets to construct the six-class ship target dataset. The number of samples for each ship target category is shown in Table I. To standardize ship chip sizes, we generate 100×100 SAR images via center cropping. In the experiments, 70% of the ship chips are allocated to the training set, while the remaining 30% are used for the test set.

B. Experimental Settings and Evaluation Metrics

In the experiments, the MATLAB platform is used for scattering center extraction and complex-valued SAR image reconstruction, and the number of extracted scattering centers is set to 40, i.e., $Q = 40$. Meanwhile, the PyTorch deep learning framework, running on an NVIDIA GeForce RTX3090 GPU with Ubuntu 20.04 Linux, is used to train EFMM-Net. During network training, a cosine annealing learning rate scheduler with the half-period of 50 epochs and the maximum learning rate of 0.001 is adopted to train the network. The periodic adjustment of learning rate helps the network to get rid of local minima and achieve precise parameter fine-tuning, thus promoting rapid and stable convergence. Furthermore, an Adam optimizer with 1e-3 weight decay is used to provide

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TABLE II
CONFUSION MATRIX OF THE THREE-CLASS SHIP TARGET DATASET

True Label	Predicted Label			F1-Score (%)
	Bulk Carrier	Container Ship	Tanker	
Bulk Carrier	160	11	6	88.89
Container Ship	12	114	7	86.69
Tanker	11	5	118	89.06
Recall (%)	90.40	85.71	88.06	Accuracy (%) = 88.29
Precision (%)	87.43	87.69	90.08	

regularization constraints for network training, thereby avoiding overfitting phenomena. We set the batch size to 128 and train the network for 300 epochs.

In this article, recall, precision, F1 score, and accuracy metrics are used to evaluate the performance of EFMM-Net, and the recognition accuracy is taken as the main evaluation metric. These metrics can be calculated according to the confusion matrix

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{F1-Score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \tag{19}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{20}$$

where TP, FP, FN, and TN denote the number of true positive, false positive, false negative, and true negative instances, respectively.

C. Recognition Results and Comparison

In this section, the experiments are conducted on the three-class and six-class SAR ship target datasets, and the recognition results of the proposed EFMM-Net method are compared with classical deep learning methods and SAR target recognition methods.

Tables II and III are the confusion matrices of the proposed EFMM-Net method on the three-class and six-class ship target datasets. As Table II shows, Bulk Carrier and Container Ship targets are structurally and dimensionally similar, making them prone to confusion. Overall, EFMM-Net demonstrates high recognition performance, achieving over 85% in recall, precision, and F1-score for three-class ship targets, with an overall accuracy of 88.29%. As Table III shows, the General Cargo target exhibits the lowest recall rate of 41.43%. This reason lies in the fact that the low resolution of the OpenSARShip 2.0 dataset and the high similarity in hull shape and cabin layout between General Cargo ships and other large ship targets. However, the total recognition accuracy of six-class ship targets reaches 69.64%, which is at a relatively high level.

To further validate the superiority of EFMM-Net method for SAR target recognition, classical deep learning methods including VGG-16 [45], ResNet-18 [46], DenseNet-121 [47], EffcientNet-V2 [48], and ViT [49] are selected and compared with the proposed method. In addition, we compare the recognition results of EFMM-Net with A-ConvNets [15] and other networks known for their good performance in ship target recognition, including HDSS-Net [50], EFA-DHFR [51], DSN [21], DBDN [22], and MGSFA-Net [27]. Among them, the A-ConvNets, HDSS-Net, and EFA-DHFR networks mainly consider the visual and textural features of SAR amplitude images, while the DSN and DBDN methods fuse the time–frequency and polarization features of the ship target, respectively. Both MGSFA-Net and the proposed EFMM-Net method incorporate the electromagnetic scattering features of the ship target. The comparison results are shown in Table IV, where the recognition results of our methods are marked in bold and the optimal results of comparison methods are underlined. The deep learning methods are mainly oriented to optical images without considering the unique characteristics of SAR targets, so the recognition performance is inferior to networks specifically designed for SAR target recognition. Among the SAR target recognition methods, the DBDN method integrates the polarization and texture features of the ship target, thereby achieving 87.62% recognition accuracy in the three-class ship target recognition task. Compared with MGSFA-Net, the proposed EFMM-Net method effectively uses the phase information of SAR images, so the recognition accuracy of the three-class ship target recognition task is improved 2.63%. Compared with all the comparison methods, EFMM-Net exhibits the excellent recognition performance. In the three-class and six-class ship target recognition tasks, it has improved 0.67% and 2.61% over the suboptimal DBDN and EFA-DHFR methods, respectively.

D. Recognition Performance Under Different Polarizations

The report of the OpenSARShip dataset shows that SAR images of the ship target acquired under different polarizations exhibit different backscattering characteristics [43], [44]. For example, the VV polarization is more sensitive to double-bounce scattering, while the VH polarization is more sensitive to volume scattering. Therefore, we further investigate the recognition performance of the proposed EFMM-Net method for ship targets under different polarizations, and the results are presented in Table V. It can be seen that the EFMM-Net method under VV polarization can always achieve better recognition performance in the three-class and six-class ship target recognition tasks, and the recognition accuracy under VV polarization is 0.83% and 2.05% higher than VH polarization, respectively. The reason lies in the fact that ship targets typically exhibit stronger backscattering values in the copolarization VV channel, making the structure of ship targets more apparent with better recognition performance. In addition, even with single-polarized data, the proposed EFMM-Net method can achieve higher recognition accuracy than the deep learning methods and A-ConvNets method, shown in Table IV. Overall, SAR images with different polarizations reflect the scattering characteristics of different structures of ship targets, so using the total SAR images with VV and VH polarizations to train our model helps learn more target structural features, thus achieving better recognition results.

E. Ablation Study

In this section, with recognition accuracy as the main evaluation metric, ablation experiments are conducted on the

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TABLE III
CONFUSION MATRIX OF THE SIX-CLASS SHIP TARGET DATASET

True Label	Predicted Label						F1-Score (%)
	Bulk Carrier	Container Ship	Tanker	Cargo	General Cargo	Fishing	
Bulk Carrier	137	14	5	13	6	1	73.46
Container Ship	13	102	6	7	3	0	77.56
Tanker	16	3	78	12	18	9	65.27
Cargo	19	4	7	99	4	6	67.35
General Cargo	10	7	5	16	29	3	44.61
Fishing	2	2	2	8	0	62	78.98
Recall (%)	77.84	77.86	57.35	71.22	41.43	81.58	Accuracy (%) = 69.64
Precision (%)	69.54	77.27	75.73	63.87	48.33	76.54	

TABLE IV
COMPARISON OF RECOGNITION RESULTS (%) OF DIFFERENT METHODS ON OPENSARSHIP 2.0 THREE-CLASS AND SIX-CLASS SHIP TARGETS

Methods		Three-Class Recognition				Six-Class Recognition			
		Recall	Precision	F1-Score	Accuracy	Recall	Precision	F1-Score	Accuracy
Deep Learning Methods	VGG-16 [45]	72.57	71.17	72.30	71.85	50.69	54.51	49.03	52.75
	ResNet-18 [46]	73.15	73.62	73.33	75.74	41.06	44.09	41.04	53.26
	DenseNet-121 [47]	74.75	72.91	72.88	74.75	40.59	53.35	41.43	51.70
	EffcientNet-V2 [48]	74.77	75.80	74.96	76.59	44.75	54.00	45.73	55.41
	Vision Transformer [49]	74.67	74.33	73.33	72.97	49.67	49.83	46.67	51.10
SAR Target Recognition Methods	A-ConvNets [15]	78.39	77.62	77.81	77.93	56.44	58.21	55.34	57.83
	HDSS-Net [50]	80.66	81.16	80.91	83.23	59.53	60.26	59.89	60.76
	EFA-DHFR [51]	85.38	85.76	85.57	85.40	65.99	68.69	67.31	67.03
	DSN [21]	82.58	82.38	82.18	82.88	51.37	54.32	51.99	60.19
	DBDN [22]	86.32	87.31	86.74	87.62	55.43	65.60	58.44	64.26
	MGSFA-Net [27]	85.97	83.86	84.90	85.66	-	-	-	-
Ours	**EFMM-Net**	**88.06**	**88.40**	**88.21**	**88.29**	**67.88**	**68.55**	**67.87**	**69.64**

TABLE V
RECOGNITION RESULTS (%) UNDER DIFFERENT POLARIZATIONS ON OPENSARSHIP 2.0 THREE-CLASS AND SIX-CLASS SHIP TARGETS

Tasks	Polarization	Recall	Precision	F1-Score	Accuracy
Three-Class	VV	80.15	81.34	80.42	80.58
	VH	80.10	79.51	79.72	79.75
	Total	**88.06**	**88.40**	**88.21**	**88.29**
Six-Class	VV	60.05	63.62	60.91	63.44
	VH	59.89	60.95	58.14	61.39
	Total	**67.88**	**68.55**	**67.87**	**69.64**

TABLE VI
RECOGNITION ACCURACY (%) OF DIFFERENT MODELS FOR THE OPENSARSHIP 2.0 THREE-CLASS AND SIX-CLASS DATASETS

Methods	Tasks	
	Three-Class	Six-Class
SSMM-Net$^{(d)}$	83.11	62.64
SSMM-Net$^{(e)}$	78.38	61.68
EFMM-Net†	87.84	68.41
EFMM-Net	**88.29**	**69.64**

OpenSARShip 2.0 three-class and six-class ship target datasets to verify the effectiveness of main modules of the proposed EFMM-Net method.

First, to validate the complementary roles of data features and electromagnetic scattering features in SAR target recognition, we conduct architectural modifications by converting the dual-stream manifold multiscale network into two single-stream variants. Among them, SSMM-Net$^{(d)}$ processes only the original complex-valued SAR images, while SSMM-Net$^{(e)}$ operates exclusively on the reconstructed complex-valued SAR images. After the input data pass through the manifold multiscale network, the extracted data features or electromagnetic scattering features are directly input into the classification head for target category prediction. The ablation experimental results in Table VI show that the proposed dual-stream EFMM-Net network exhibits better recognition accuracy than the two single-stream networks. Furthermore, we conduct additional ablation study on the proposed scattering-guided learning method. EFMM-Net† denotes the SGMM backbone without channelwise feature alignment, i.e., the auxiliary loss term L_{CN} is ablated. Compared with EFMM-Net†, our EFMM-Net method achieves consistently higher accuracy on both the tasks, demonstrating the effectiveness of scattering-guided channelwise feature alignment.

Then, the manifold feature extraction part within the manifold multiscale network is investigated. Without changing the network structure of multiscale feature extraction and fusion part, we replace the manifold network with the real-valued convolution. Specifically, the wFM convolution is changed to the traditional real-valued convolution with ReLU activation function, the manifold FC layer is removed, and the dual-stream input data are replaced with SAR amplitude images.

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TABLE VII
ABLATION EXPERIMENTS OF THE PROPOSED MANIFOLD FEATURE EXTRACTION PART

Tasks	Models		Accuracy (%)
	Real-Valued Network	Manifold Network	
Three-Class	✓	×	84.91
	×	✓	**88.29**
Six-Class	✓	×	65.38
	×	✓	**69.64**

TABLE VIII
ABLATION EXPERIMENTS OF MULTISCALE FEATURE EXTRACTION AND FUSION PART

Tasks	Models			Accuracy (%)
	Branch1	Branch2	Branch3	
Three-Class	✓	×	×	81.53
	×	✓	×	82.21
	×	×	✓	83.78
	✓	✓	✓	**88.29**
Six-Class	✓	×	×	66.89
	×	✓	×	68.13
	×	×	✓	67.99
	✓	✓	✓	**69.64**

TABLE IX
ABLATION EXPERIMENTS OF THE PROPOSED LAFF METHOD

Tasks	Fusion Methods	Accuracy (%)
Three-Class	Concatenation	82.43
	Gated Unit	84.46
	LAFF	**88.29**
Six-Class	Concatenation	66.07
	Gated Unit	67.31
	LAFF	**69.64**

As Table VII shows, EFMM-Net demonstrates superior recognition performance on both the three-class and six-class datasets compared with the real-valued variant. We can come to the conclusion that the manifold network can fully use the phase information of complex-valued SAR image to improve the target recognition accuracy.

To evaluate the effectiveness of multiscale feature extraction and fusion part, we conduct independent experiments for the three scale branches, and Table VIII shows the experimental results. It can be observed that since Bulk Carrier, Container Ship, and Tanker targets usually have large sizes, relatively high recognition accuracy can be obtained by using the scale branch 3 in the three-class ship target recognition task. For the six-class target recognition task, higher recognition accuracy can be obtained using the scale branch 2 due to the presence of small fishing vessel targets. However, our method achieves the highest recognition accuracy, reaching 88.29% and 69.64% on the three-class and six-class datasets, respectively. The experimental results further suggest that the multiscale feature extraction and fusion part can effectively enhance the feature representation of the ship target, thus improving the recognition ability of the model.

TABLE X
RECOGNITION ACCURACY (%) OF THREE-CLASS SHIP TARGET UNDER DIFFERENT NUMBERS OF TRAINING SAMPLES

Class	Training Number in Each Class				
	20	40	60	80	100
Bulk Carrier	74.58	79.66	81.36	83.62	82.49
Container Ship	57.14	63.91	64.66	70.68	80.45
Tanker	67.91	76.12	77.61	83.58	82.09
Average	**67.34**	**73.87**	**75.23**	**79.73**	**81.76**

TABLE XI
RECOGNITION ACCURACY (%) OF SIX-CLASS SHIP TARGET UNDER DIFFERENT NUMBERS OF TRAINING SAMPLES

Class	Training Number in Each Class				
	20	40	60	80	100
Bulk Carrier	55.68	60.23	62.50	69.32	70.45
Container Ship	62.59	66.41	63.36	60.31	64.89
Tanker	40.44	46.32	57.35	53.68	58.82
Cargo	49.64	56.83	52.52	60.43	57.55
General Cargo	37.14	38.57	35.71	41.43	42.86
Fishing	71.05	71.05	77.63	81.58	78.95
Average	**52.75**	**57.14**	**60.16**	**61.68**	**63.05**

Finally, we compare the LAFF with two typical feature fusion methods to verify the effectiveness of the proposed fusion method. The first method concatenates the data features and electromagnetic scattering features along the channel dimension and then feeds the fusion features into the classification head for target category prediction. The second method applies a gated multimodal unit to fuse these two features [52]. Table IX reports the experimental results of different feature fusion methods. It is evident that the LAFF is superior to the other two comparison methods, especially on the three-class ship target recognition task. The recognition accuracy of LAFF is improved by 5.86% over the concatenation fusion method, and by 3.83% over the gated unit fusion method. The main reason is that LAFF can obtain the spatial location information of the target through coordinate attention, thus promoting the model to focus more on the target region and improving the SAR target recognition accuracy.

F. Recognition Performance Under Complex Conditions

1) Limited Training Sample Condition: In practical applications, the training samples are usually insufficient. In this section, the recognition performance of EFMM-Net with limited training samples is evaluated on the OpenSARShip 2.0 three-class and six-class ship target datasets. During the experiments, we first fix the random seed and then randomly select 20, 40, 60, 80, and 100 training samples from each ship target category, and the remaining samples are used for testing.

The recognition accuracies of three-class and six-class ship targets are shown in Tables X and XI, respectively. It can be seen that with the training samples of each ship target category set to 100, the recognition accuracy reaches 81.76% for three-class ship targets and 63.05% for six-class ship targets, surpassing those of all the deep learning methods and A-ConvNets baseline method shown in Table IV. In addition,

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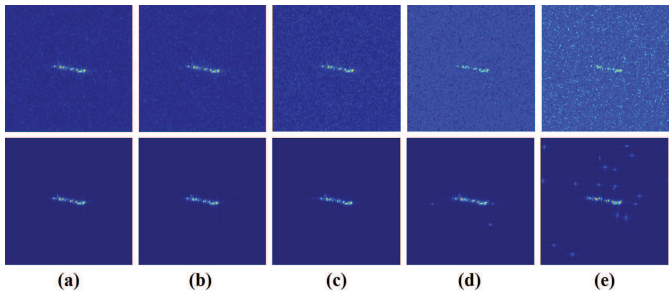


Fig. 8. SAR images of the bulk carrier target under different SNRs. The first row shows the original SAR images disturbed by noise, and the second row shows the reconstructed SAR images. (a) 10 dB. (b) 5 dB. (c) 0 dB. (d) −5 dB. (e) −10 dB.

even if the number of training samples per category is reduced to 40, the recognition accuracy of the three-class and six-class ship target recognition tasks drops by up to approximately 8% at most, which indicates that our method has strong robustness under limited training sample condition.

2) Noise Interference Condition: During the imaging process, the noise caused by the radar system and background will affect the quality of SAR images. In severe cases, the target may be submerged by noise, thus making SAR target recognition more difficult. To simulate the noise interference, we manually add the additive complex white Gaussian noise to the original complex-valued SAR data and construct the three-class ship target datasets with signal-to-noise ratios (SNRs) of 10, 5, 0, −5, and −10 dB, respectively. Fig. 8 shows the original and the reconstructed SAR images of the Bulk Carrier target under different SNRs. From left to right are the SAR images with the SNR of 10, 5, 0, −5 and −10 dB, respectively, and the first and second rows correspond to the original and reconstructed SAR images, respectively. It can be seen that with the decrease in SNR, the noise interference to the original SAR images is gradually aggravated, while the reconstructed SAR images show strong robustness to noise interference. With the SNR set to −10 dB, it is difficult to distinguish the contour of ship target from the original SAR image. However, in the reconstructed SAR image, most of the background noise is filtered out, and the contour of the ship target is relatively clear.

Fig. 9 shows the variation folds of the recognition accuracy of A-ConvNets, SSMM-Net$^{(d)}$, SSMM-Net$^{(e)}$, EFMM-Net†, and the proposed EFMM-Net method for three-class ship targets under different SNRs. As the SNR decreases from 10 to −10 dB, the A-ConvNets method is the most sensitive to noise interference and has the largest drop in recognition accuracy, followed by the SSMM-Net$^{(d)}$ method. In contrast, the SSMM-Net$^{(e)}$, EFMM-Net†, and EFMM-Net methods use the electromagnetic scattering features with stronger robustness to noise interference, so their recognition accuracy drops more smoothly as the SNR decreases. The recognition accuracy of EFMM-Net always outperforms that of EFMM-Net† method under all the noise levels, demonstrating that the proposed scattering-guided channelwise feature alignment method can further enhance ship target recognition robustness under noise interference. Even if the SNR is reduced to −10 dB, the proposed EFMM-Net method can still achieve more than 80%

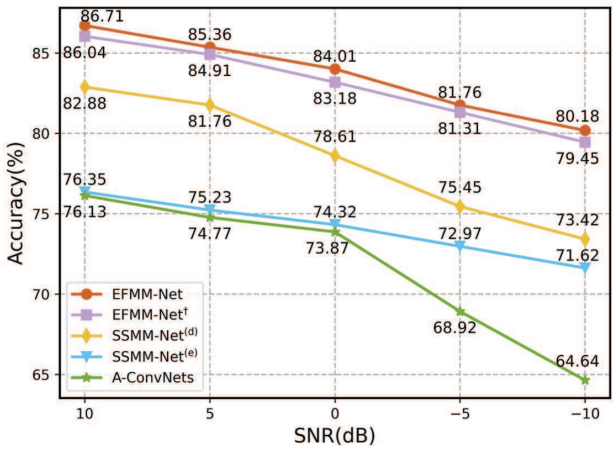


Fig. 9. Recognition accuracy of different methods for three-class ship targets under different SNRs.

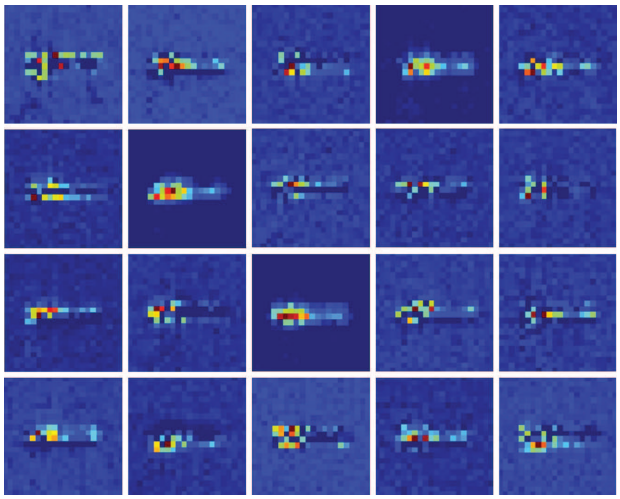


Fig. 10. Manifold feature maps of original complex-valued SAR image.

recognition accuracy in the three-class ship target recognition task, exhibiting strong anti-noise interference capability.

G. Effectiveness of Scattering-Guided Learning

In this section, combined with the visualization methods, the effectiveness of scattering-guided learning method is analyzed in detail. The EFMM-Net learns the features of each sample during SAR target recognition, so it is easier to observe the role of electromagnetic scattering features for SAR target recognition by visualizing the intermediate features of a single sample. Figs. 10 and 11 show the manifold feature maps of the original and the reconstructed complex-valued SAR images, respectively. As shown in Fig. 10, most of the feature channels learn both the target and background region features, and only three channels purely learn the target region features. In contrast, in the manifold feature maps of the reconstructed complex-valued SAR image shown as Fig. 11, all the feature channels are used to learn the target region features, exhibiting strong target attention. Fig. 12 further presents the manifold feature maps of original complex-valued SAR image after feature alignment. The channelwise feature alignment drives the feature maps of original complex-valued SAR image

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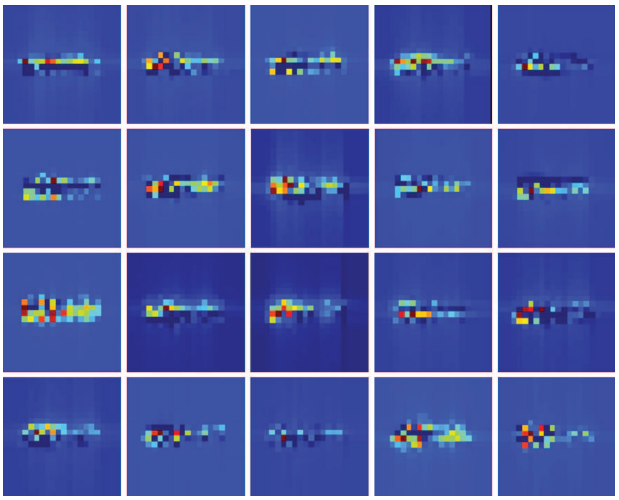


Fig. 11. Manifold feature maps of reconstructed complex-valued SAR image.

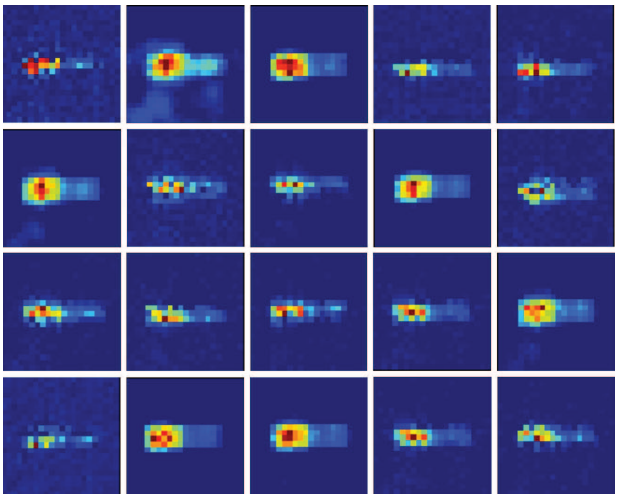


Fig. 12. Manifold feature maps of original complex-valued SAR image after feature alignment.

to approach those of the reconstructed one. As a result, the aligned manifold features significantly enhance structural scattering features in target regions while retaining the useful target information for recognition.

We further visualize the attention heat maps of the single-stream SSMM-Net$^{(d)}$ and the proposed EFMM-Net methods using the Grad-CAM method [53]. The Grad-CAM method can determine spatial locations of interest through gradient backpropagation, thereby highlighting regions considered important by the model. As seen in Fig. 13, the first row corresponds to the original SAR images of ship targets. The second and third rows are the Grad-CAM heat maps of SSMM-Net$^{(d)}$ and EFMM-Net methods for ship targets, respectively. It can be seen that compared with the SSMM-Net$^{(d)}$ method, the heat maps of the EFMM-Net method focus mainly on the target region rather than the background region. It can be demonstrated that the electromagnetic scattering features can direct the recognition process of EFMM-Net, so as to improve the SAR target recognition performance.

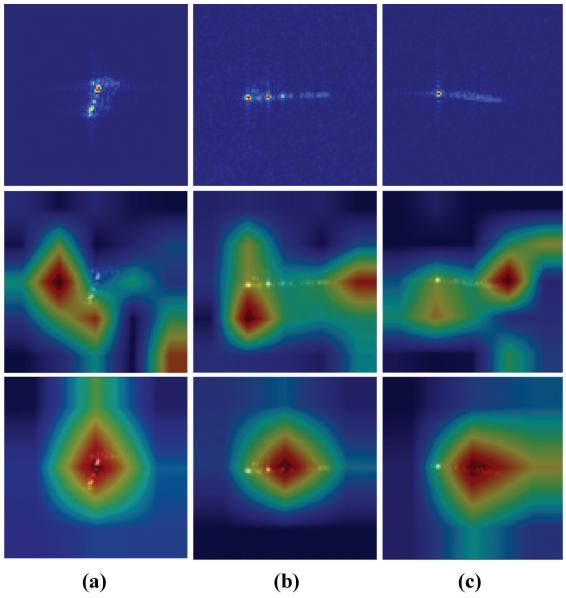


Fig. 13. Grad-CAM heat maps of (a) Bulk carrier, (b) Container ship, and (c) Tanker. From top to bottom, the first row corresponds to the original SAR images, and the second and third rows correspond to the heat maps of SSMM-Net$^{(d)}$ and EFMM-Net methods, respectively.

V. Conclusion

To achieve high-precision and robust SAR target recognition, this article proposes a novel EFMM-Net method. Specifically, EFMM-Net adopts the original complex-valued SAR image and the reconstructed complex-valued SAR image using the ASC model as dual-stream input. Then, the SGMM backbone is proposed to perform parallel extraction of both the data features and electromagnetic scattering features of the target. Within the SGMM backbone, a scattering-guided learning method is introduced to dynamically adjust data features through channelwise feature alignment, thereby enhancing the awareness of data features to critical scattering characteristics. Finally, the LAFF recognition module realizes effective feature fusion and SAR target recognition. Compared with previous SAR target recognition methods, EFMM-Net effectively uses the phase information and preserves the geometric properties of the complex data, while integrating the data features and electromagnetic scattering characteristics within a unified framework. The experimental results on the OpenSARShip 2.0 dataset demonstrate that the proposed EFMM-Net method outperforms current classical deep learning baselines and specialized SAR target recognition approaches in recognition accuracy.

In the future, we aim to further investigate the SAR target recognition methods combining the full polarization scattering characteristics of the target. Meanwhile, we will explore the integration of SAR imaging and recognition in combination with deep learning imaging techniques.

References

[1] G. Xu, M. Xing, L. Zhang, and Z. Bao, "Robust autofocusing approach for highly squinted SAR imagery using the extended wavenumber algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 10, pp. 5031–5046, Oct. 2013.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 07:00:57 UTC from IEEE Xplore. Restrictions apply.

[2] B. Zhang, G. Xu, R. Zhou, H. Zhang, and W. Hong, "Multi-channel back-projection algorithm for mmWave automotive MIMO SAR imaging with Doppler-division multiplexing," *IEEE J. Sel. Topics Signal Process.*, vol. 17, no. 2, pp. 445–457, Mar. 2023.

[3] H. Lin, H. Wang, F. Xu, and Y.-Q. Jin, "Target recognition for SAR images enhanced by polarimetric information," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5204516.

[4] G. Xu, B. Zhang, H. Yu, J. Chen, M. Xing, and W. Hong, "Sparse synthetic aperture radar imaging from compressed sensing and machine learning: Theories, applications, and trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 4, pp. 32–69, Dec. 2022.

[5] Z. Zhou, L. Zhao, K. Ji, and G. Kuang, "A domain-adaptive few-shot SAR ship detection algorithm driven by the latent similarity between optical and SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5216318.

[6] H. Jia, X. Pu, Q. Liu, H. Wang, and F. Xu, "A fast progressive ship detection method for very large full-scene SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5206615.

[7] J. Geng, S. Song, and W. Jiang, "Dual-path feature aware network for remote sensing image semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3674–3686, May 2024.

[8] K. Wu, F. Cai, and H. Wang, "CTMANet: A CNN-transformer hybrid semantic segmentation network for fine-grained airport extraction in complex SAR scenes," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4689–4704, 2024.

[9] Z. Wei, W. Jie, and G. Jian, "An efficient SAR target recognition algorithm based on contour and shape context," in *Proc. 3rd Int. Asia Pacific Conf. Synth. Aperture Radar (APSAR)*, Sep. 2011, pp. 1–4.

[10] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Diego, CA, USA, Jun. 2005, pp. 886–893.

[11] A. Hyvrinen, J. Karhunen, and E. Oja, *Independent Component Analysis*. Hoboken, NJ, USA: Wiley, 2004.

[12] A. K. Mishra, "Validation of PCA and LDA for SAR ATR," in *Proc. IEEE Region Conf.*, Hyderabad, India, Nov. 2008, pp. 1–6.

[13] B. Ding, G. Wen, J. Zhong, C. Ma, and X. Yang, "A robust similarity measure for attributed scattering center sets with application to SAR ATR," *Neurocomputing*, vol. 219, pp. 130–143, Jan. 2017.

[14] P. Ni, Y. Liu, H. Pei, H. Du, H. Li, and G. Xu, "CLISAR-net: A deformation-robust ISAR image classification network using contrastive learning," *Remote Sens.*, vol. 15, no. 1, p. 33, Dec. 2022.

[15] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[16] H. Pei, M. Su, G. Xu, M. Xing, and W. Hong, "Self-supervised feature representation for SAR image target classification using contrastive learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9246–9258, 2023.

[17] J. He, Y. Wang, and H. Liu, "Ship classification in medium-resolution SAR images via densely connected triplet CNNs integrating Fisher discrimination regularized metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3022–3039, Apr. 2021.

[18] X. Bai, R. Xue, L. Wang, and F. Zhou, "Sequence SAR image classification based on bidirectional convolution-recurrent network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9223–9235, Nov. 2019.

[19] H. Dong, L. Zhang, and B. Zou, "Exploring vision transformers for polarimetric SAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219715.

[20] H. Ren et al., "Transductive prototypical attention reasoning network for few-shot SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206813.

[21] Z. Huang, M. Datcu, Z. Pan, and B. Lei, "Deep SAR-net: Learning objects from signals," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 179–193, Mar. 2020.

[22] N. Xie, T. Zhang, W. Guo, Z. Zhang, and W. Yu, "Dual branch deep network for ship classification of dual-polarized SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5207415.

[23] J. Oh, G.-Y. Youm, and M. Kim, "SPAM-net: A CNN-based SAR target recognition network with pose angle marginalization learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 2, pp. 701–714, Feb. 2021.

[24] R. Wang, T. Su, D. Xu, J. Chen, and Y. Liang, "MIGA-net: Multi-view image information learning based on graph attention network for SAR target recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 10779–10792, Nov. 2024.

[25] J. Zhang, M. Xing, and Y. Xie, "FEC: A feature fusion framework for SAR target recognition based on electromagnetic scattering features and deep CNN features," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2174–2187, Mar. 2021.

[26] Y. Li, L. Du, and D. Wei, "Multiscale CNN based on component analysis for SAR ATR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5211212.

[27] X. Zhang, S. Feng, C. Zhao, Z. Sun, S. Zhang, and K. Ji, "MGSFA-net: Multiscale global scattering feature association network for SAR ship target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4611–4625, 2024.

[28] Z. Liu, L. Wang, Z. Wen, K. Li, and Q. Pan, "Multilevel scattering center and deep feature fusion learning framework for SAR target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5227914.

[29] S. Feng, K. Ji, L. Zhang, X. Ma, and G. Kuang, "SAR target classification based on integration of ASC parts model and deep learning algorithm," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10213–10225, 2021.

[30] S. Feng, K. Ji, F. Wang, L. Zhang, X. Ma, and G. Kuang, "PAN: Part attention network integrating electromagnetic characteristics for interpretable SAR vehicle target recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5204617.

[31] L. Liao, L. Du, J. Chen, Z. Cao, and K. Zhou, "'EMI-Net: An end-to-end mechanism-driven interpretable network for SAR target recognition under EOCs," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5205118.

[32] Z. Zeng, J. Sun, Z. Han, and W. Hong, "SAR automatic target recognition method based on multi-stream complex-valued networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5228618.

[33] Y. Zhu, T. Li, D. Peng, H. Wang, and S. Shi, "A novel SAR automatic target recognition method based on fully complex-valued networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6160–6171, 2023.

[34] R. Chakraborty, Y. Xing, and S. X. Yu, "SurReal: Complex-valued learning as principled transformations on a scaling and rotation manifold," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 940–951, Mar. 2022.

[35] R. Chakraborty, J. Bouza, J. H. Manton, and B. C. Vemuri, "ManifoldNet: A deep neural network for manifold-valued data with applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 799–810, Feb. 2022.

[36] Q. Liu and L. Lang, "MMFF: Multi-manifold feature fusion based neural networks for target recognition in complex-valued SAR imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 151–162, Oct. 2021.

[37] J. Duan, L. Zhang, Y. Wu, M. Xing, and M. Wu, "A novel signal reconstructing method for radar targets," in *Proc. IEEE China Summit Int. Conf. Signal Inf. Process. (ChinaSIP)*, Jul. 2014, pp. 175–178.

[38] Y. Fan, B. Zhang, and Y. Wu, "Integrating regularization and PnP priors for SAR image reconstruction using multiagent consensus equilibrium," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5228115.

[39] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021.

[40] G. Wang, N. Zhang, J. Wang, W. Liu, Y. Xie, and H. Chen, "Knowledge distillation-based lightweight change detection in high-resolution remote sensing imagery for on-board processing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3860–3877, 2024.

[41] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.

[42] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13713–13722.

[43] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[44] B. Li, B. Liu, L. Huang, W. Guo, Z. Zhang, and W. Yu, "OpenSARShip 2.0: A large-volume dataset for deeper interpretation of ship targets in Sentinel-1 imagery," in *Proc. SAR Big Data Era: Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–5.

[45] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd Intl. Conf. Learning Represent. (ICLR)*, 2015, pp. 1–14.

[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[47] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

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[48] M. Tan and Q. V. Le, "EfficientNetv2: Smaller models and faster training," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10096–10106.
[49] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Jun. 2021, pp. 1–22.
[50] Y. Shang et al., "HDSS-Net: A novel hierarchically designed network with spherical space classifier for ship recognition in SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5222420.
[51] C. Wang, S. Luo, Y. Huang, J. Pei, Y. Zhang, and J. Yang, "SAR ATR method with limited training data via an embedded feature augmenter and dynamic hierarchical-feature refiner," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5216215.
[52] J. Arevalo, T. Solorio, M. Montes-Y-Gómez, and F. A. González, "Gated multimodal units for information fusion," 2017, *arXiv:1702.01992*.
[53] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 618–626.



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